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10 AI UX principles for designing great AI experiences

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For the past 3 years, AI has been dominating a lot of conversations around digital products. In this context, there are some UX principles to follow to make sure we're designing good AI experiences for our users,

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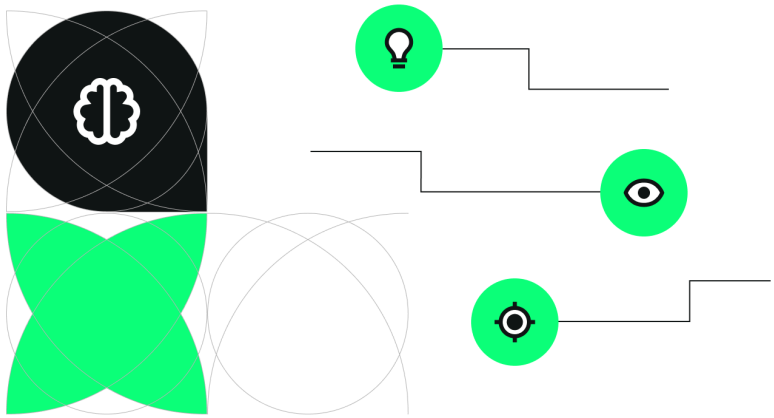

So, what should we keep in mind when designing AI products? These are powerful technologies and they can bring a lot of value to users. However, as with each new technology**, there's an initial stage when the technological development takes over, preceding any considerations about how to create a good user experience.** Understanding and testing the capabilities of the new technology seems more interesting and more captivating than putting the user at the centre.

As Jacob Nielsen highlighted this quite a few times, the big risk is that **these products become unusable in the long run**. Additionally, they could provide a very bad user experience, at the same time preventing

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AI interaction need to be established, there are still some **UX principles** that can be followed to build a good experience for AI products. These principles go beyond adding the magic icon in your products and serve as a guide for building great AI experiences.

Important to note, these AI UX principles can be applied to any AI product.

How to design AI products

1. Identify a real user need

Right now, there's a goldrush to integrate AI, especially generative AI into a lot of products. However, this can result in **overblown products**:

> *using a technology not because it's actually needed, but because it's trendy.*

While AI has developed a lot in the past two years, it still comes with challenges.

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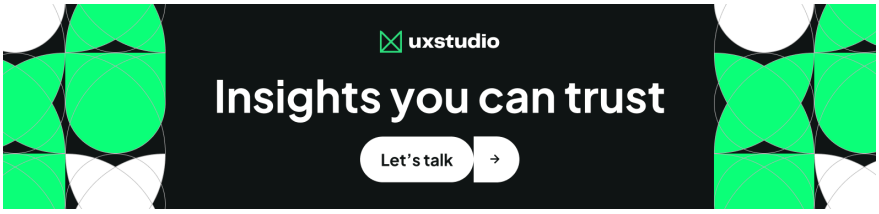
Also, probably some problems could be solved more easily without AI. AI requires constant maintenance and support, especially as things change very fast.

To make sure you create a good AI UX experience, **start by identifying the need you want the AI to solve**. Otherwise, you're giving your users tools and features that might not be that relevant for them.

The Google PAIR guidebook mentions 2 cases when AI can be introduced in a product. This is either to **automate repetitive tasks** or to **augment**, giving users super powers to do things they previously weren't able to. We won't go into details about how to make the difference between the two, but connecting the AI to a real user need and problem is a key AI UX principle.



2. Scope the AI into smaller bites

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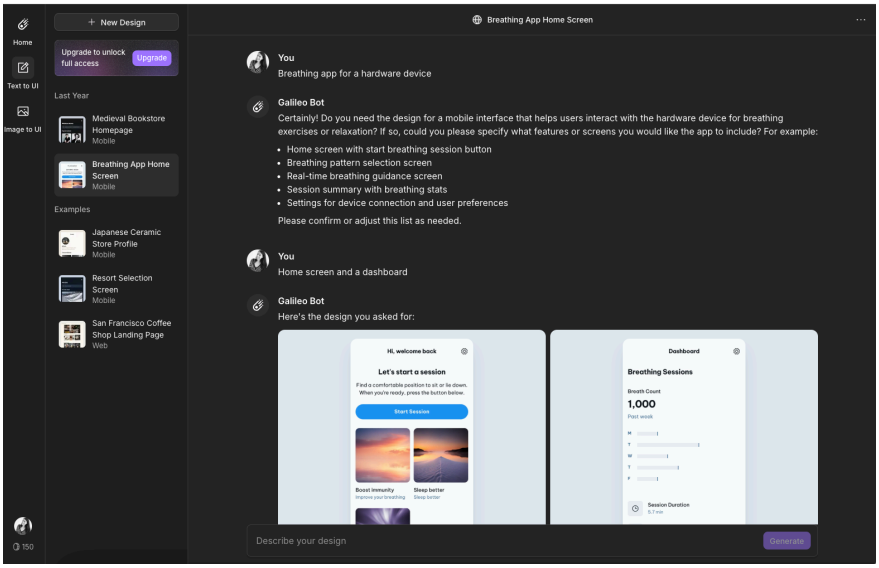
need inside your product, it's time to apply

another AI UX principle: **breaking down ambitious features into manageable tasks.** The thing is, the broader the scope of the AI, the more complicated it will be to actually build it in a way that works.



One example here would be Galileo. The platform breaks down UI generation by asking the user to narrow down specific screens or asks further details from the user in order to generate visuals that are closer to what the user needs.

We've also seen this while designing a personal finance assistant chatbot. While the chatbot was capable of answering any question from spending habits to long term

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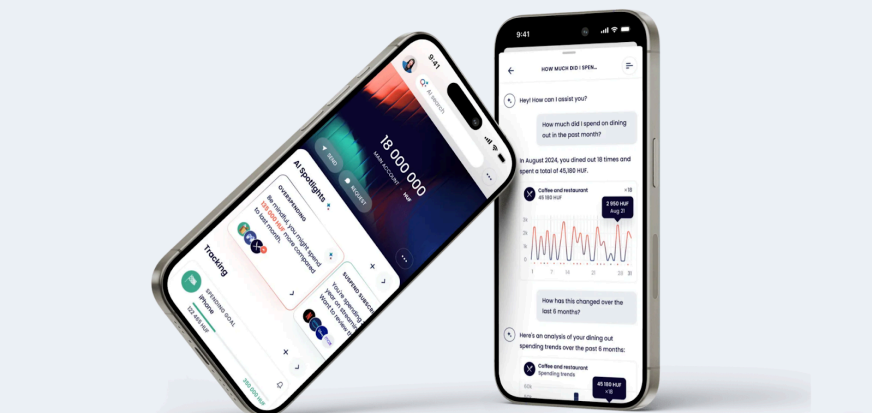

Read the case study here

Also, once you've outlined the tasks where you'll use AI to either automate and amplify human abilities, it's important to understand the users' mental models and build the AI features around them.

3. Set clear expectations for what the AI can and cannot do

AI, and especially generative AI, is a technology that can do everything and almost anything. So it's essential to **communicate clearly what it can do inside your product** or when it comes to a specific feature. Users need to know upfront what are the possibilities and what are the limitations. This is very important as it helps users understand what they can actually get from using that AI

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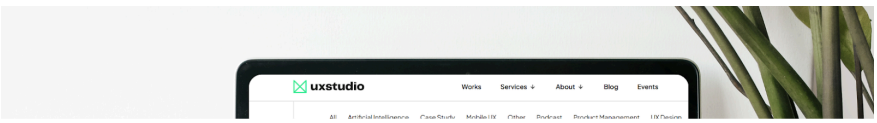
By communicating clearly what AI is capable of within your product, you **set the right expectations** and **create trust** with your users. Additionally, users won't expect things that are not currently available in your product and they won't overly depend on your product.

It should also go without saying that for any claim you make, you need to **make sure that your product delivers** on it. There are heaps of AI products making bold claims, but where the output has a lot of room for improvement, they fall short of the promise they previously set.



We explored the capabilities and shortcomings of AI tools in usability testing

In setting clear expectations, you can also reinforce what users are not allowed to do in

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you'll need additional data for the AI to provide more meaningful value for your users. And when it comes to data and AI, the consensus is garbage in, garbage out.

That shapes another AI UX principle: make sure that you're collecting relevant, high-quality data in order to make your AI actually work for your users. Also, **be transparent about what data you collect and how you use it.** It should go without saying, but don't collect data, especially personal and sensitive data, without your users' permission.

A clear example of that is how OpenAI came heavily under scrutiny for using copyrighted texts to train their models. While in their case there are issues related to copyrights, things can only get more complicated when it comes to personal data collection.

Data collection is such a big topic when it comes to AI, you can probably talk about it for ages. **Bias and representativeness** in data alone are very important and complex topics, but we won't go into that for now. Google's

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Machines are weird. They have their own way of working and with AI, even the most experienced engineers are left guessing as to why machines behave the way they do sometimes.

One such example is an image recognition algorithm built to recognise sheep. However, when the input was a sheep in a car, the algorithm failed to recognise the sheep. Moreover, it started identifying grass in other images as sheep. Ultimately, the researchers working on this realised that the data set used during the training contained mostly sheep surrounded by grass and the AI learned to identify the grass as sheep.

6. Proactively manage errors

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In these situations, errors are not a maybe, they're just a matter of time, size and impact.

To add to that, **large language models are especially prone to hallucinations**, where they can famously suggest adding glue as a pizza topic. Moreover, there's usually some **hidden bias** in the training data that doesn't show up during testing and is not acceptable for users.

All these situations bring us to the next AI-UX principle, which is about **proactively managing errors.** But how can that happen, if the systems are non-deterministic?

While it's hard to predict everything that can go wrong with AI, there can be **rules and guidelines** in place for then the AI output provides a weak or an incorrect output. One example in this case is the confidence interval provided in some image recognition applications. Users can see the **confidence interval** there and decide what to do next.

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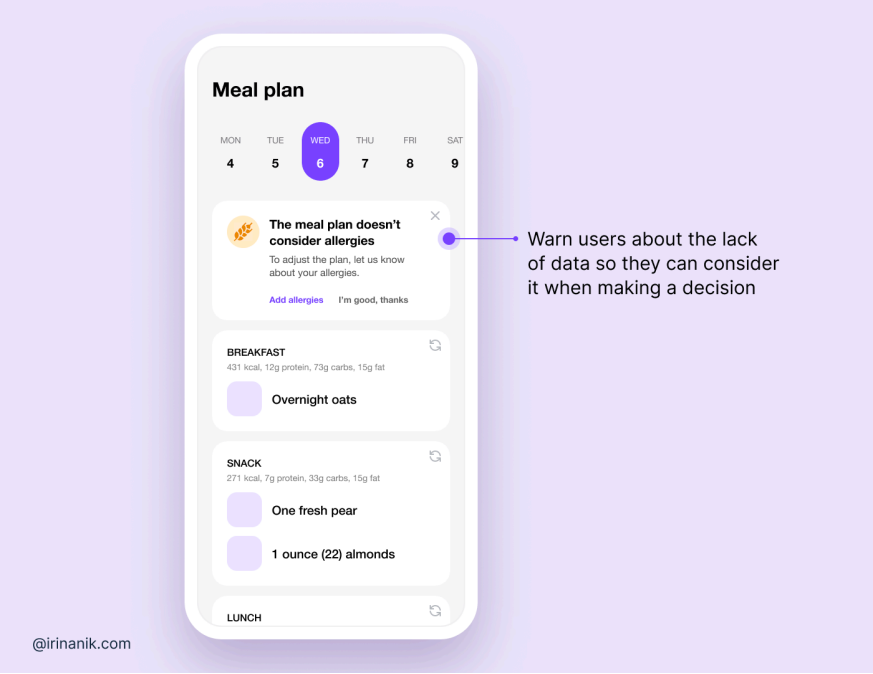

Example by Irina Nik

Managing errors is not only about coming up with solutions for solving the situation and keeping user trust when this happens. It's also about thinking about the processes or the guidelines you need to have in place in order to avoid those situations as much as possible.

To manage delicate situations, Google uses the interaction design policies concept which help develop clear guidelines for when AI systems are faulty.

Additionally, it's also about providing users with the tools to move forward. Which brings

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input, the user still needs to be given a certain amount of control over the AI.

This applies to any AI product, whether it's one that uses mostly automation and where little intervention is needed from the user once everything's set up or whether it's an agentive product that requires occasional user input.

Nest thermostats are a fitting example for this

Beyond the need to intervene whenever needed, **this AI UX principle is also sometimes referred to as the human in the loop**. Even if AI is capable of a lot of tasks of increasing complexity, human intervention is still often needed to supervise and make sure things are running smoothly, or that the

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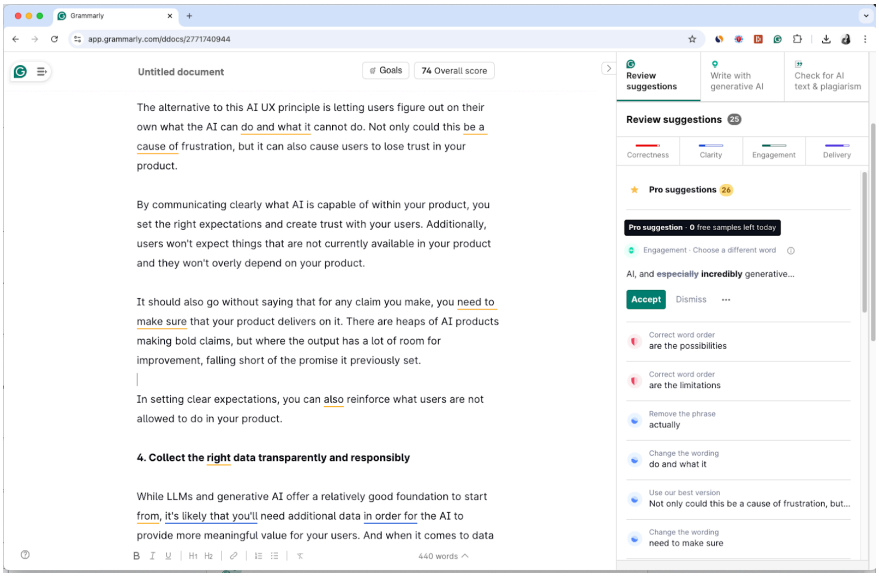
detail before and Midjourney also provides quite a few possibilities regarding this.

Grammarly is another great example. While it provides a lot of suggestions as to how users can improve their writing, **it's up to the user to decide** which suggestions they accept and which to ignore.



8. Use clear and simple language

Trust is of the essence and determines our next AI UX principle: **use clear and simple language.** Whether it's for setting expectations, error messages or communicating about how the AI actually works, it's essential that you use easy and

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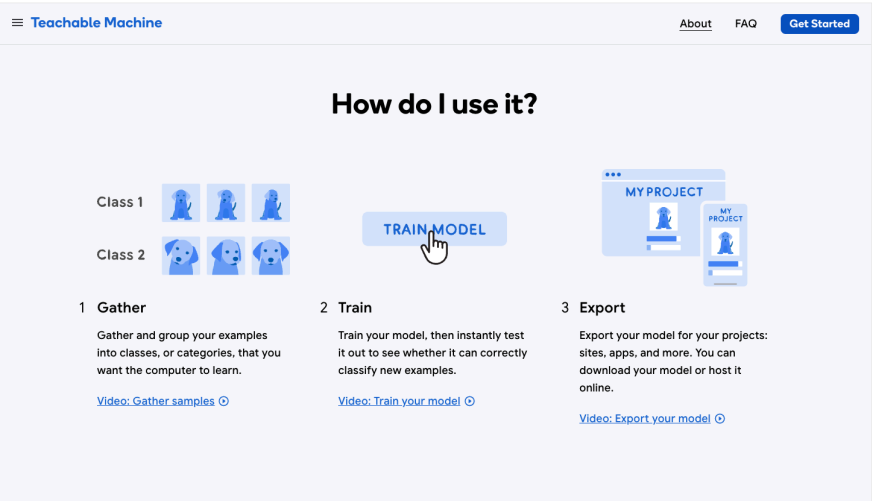
Using jargon and domain terminology is a no-no generally in UX design. In AI products, it's even more important though.

> *New technologies mean new ways of interaction. The more foreign something sounds, the harder it will be to get users started with it.*

Using clear and simple language, along with reliable outcomes, will help users build trust in your product.



Google's Teachable machine is a great example of explaining very simply how to train your own model using their product.

9. Collect user feedback

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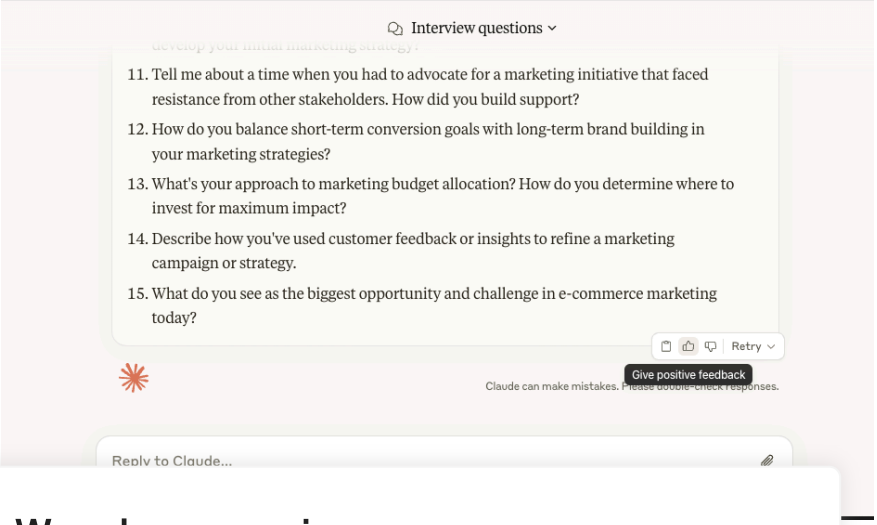
This makes collecting user feedback another important AI UX principle. There are multiple ways in which this can be done within a product, either implicitly or explicitly.

Spotify Discovery is a good example of collecting implicit feedback. When you skip one of the recommended songs, it's a signal for the platform that that particular tune was not to your liking.

A good example of explicit user feedback is asking the users to rate a specific output. While you can ask users to rate a recommendation from 1 to 10, for example. ChatGPT, as well as Claude have the thumbs up and the thumbs down so you can rate its output.



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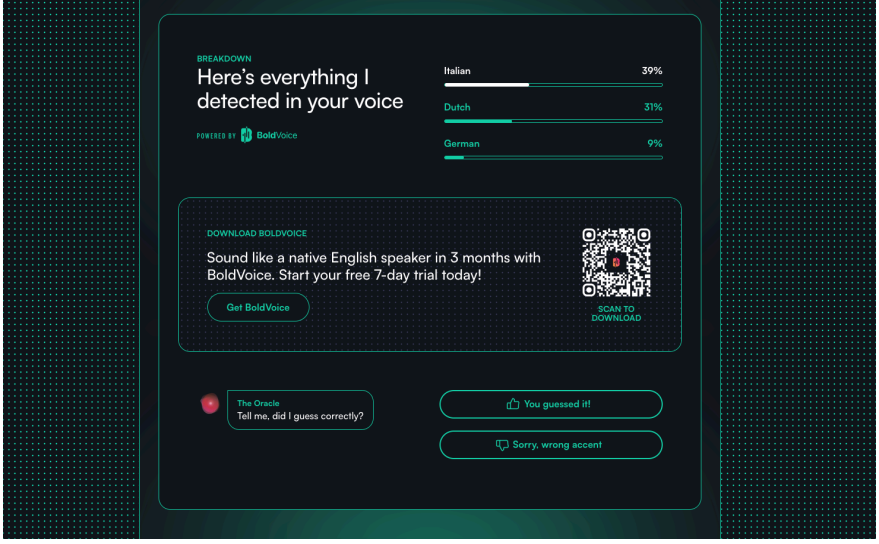

10. Make it safe & easy to try

Even if it's the last AI-UX principle on our list, it doesn't mean that it's any less important. **Any good AI product makes it easy and safe for users to experiment**. Within any product, users should be able to try it out to see its capabilities without having to worry that it will cause some unrepairable damage either to their work or their data.

Part of making it safe and easy for users to try AI in your product includes making the AI features available in multiple places within your product, where it makes sense. Here, content is key and the last thing you want to do is push users to try a feature at the wrong

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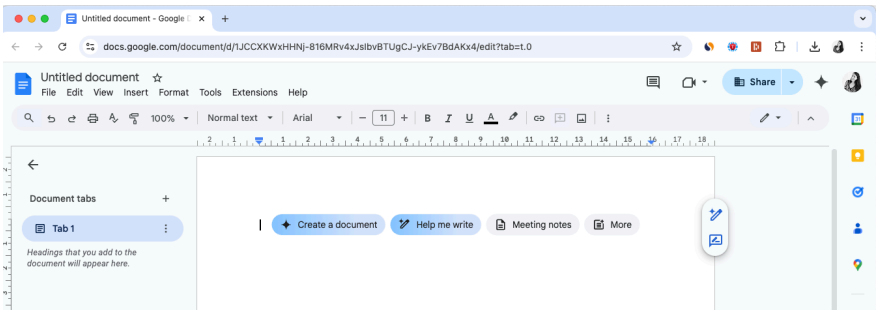
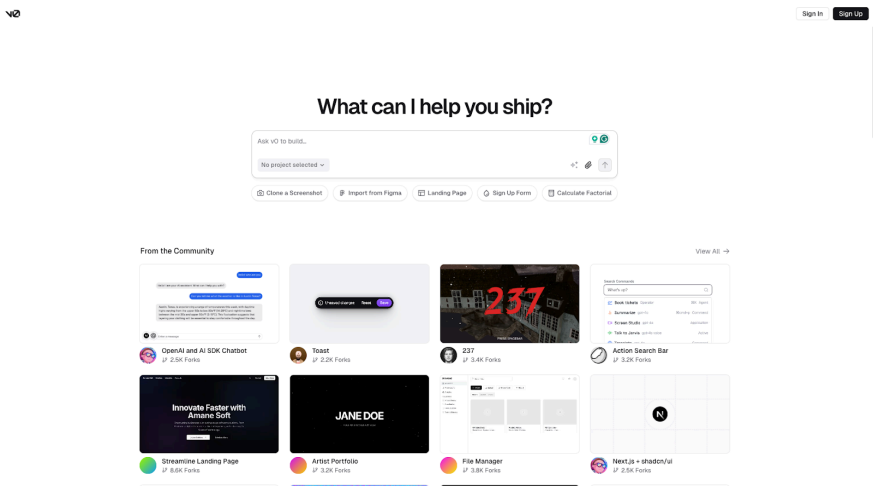


Another great example of making it safe and easy to try is V0. It allows you to test the product even without creating an account.



To summarise

In terms of UX principles, we'll probably see a lot of changes over the next few years. Especially as users will become increasingly used to multimodal, a lot of new interaction patterns will evolve, shaped by user needs and requirements. However, we believe that some

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2. Scope the AI into smaller bites

3. Set clear expectations for what the AI can and cannot do
4. Collect the right data transparently and responsibly
5. Tell users how it works – whenever possible
6. Proactively manage errors
7. Use clear and simple language
8. Give the user control
9. Collect user feedback
10. Make it safe & easy to experiment

Further resources:

- Google PAIR guidebook
- Microsoft HAX guidelines

Need some help with your AI product?

UX studio has successfully partnered with over 250 companies worldwide. If there's a challenge we can help you tackle, **get in touch with us** to discuss your AI UX strategy, or UX/UI

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